Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $(x-iy)^{\frac{1}{3}}=a-ib$, then $\frac{x}{a}+\frac{y}{b}=K\left(a^2-b^2\right)$, then the value of K is

(A) 1

(B) 2

(C) 4

(D) 3

Q.2

If $x+iy=\frac{1}{1+\cos\theta+i\sin\theta}$, then $4x^2-1=$

(A) -1

(B) 2

(C) 0

(D) 4

Q.3

If $z=2-3i$, then $z^2-4z+13=$

(A) 1

(B) 0

(C) 2

(D) -3

Q.4

The value of $\left(\frac{1+i}{1-i}\right)^{1001}+\left(\frac{1-i}{1+i}\right)^{1001}$ is

(A) 0

(B) $2^{1001}(1+\mathrm{i})$

(C) $2i$

(D) 2

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Q.5

The real value of α for which the expression $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely real is

(A) $(n+1)\frac{\pi}{2}$

(B) $(2n+1)\frac{\pi}{2}$

(C) $n\pi$

(D) none of these

Q.6

If α, β, γ and a, b, c are complex numbers such that $\frac{\alpha}{\mathrm{a}}+\frac{\beta}{\mathrm{b}}+\frac{\gamma}{\mathrm{c}}=1+\mathrm{i}$ and $\frac{\mathrm{a}}{\alpha}+\frac{\mathrm{b}}{\beta}+\frac{\mathrm{c}}{\gamma}=0$, then the value of $\frac{\alpha^2}{\mathrm{a}^2}+\frac{\beta^2}{\mathrm{b}^2}+\frac{\gamma^2}{\mathrm{c}^2}$ is equal to

(A) 0

(B) -1

(C) $2\mathrm{i}$

(D) $-2\mathrm{i}$

Q.7

If complex number z satisfies $|z-6i| = \mathrm{lm}(z)$, then range of $(\mathrm{argz}-\mathrm{argz}^{-})$ will be

(A) $[\pi/2, 3\pi/2]$

(B) $[2\pi/4\pi/3]$

(C) $[\pi/3, 2\pi/3]$

(D) $[3\pi/4, 5\pi/3]$

Q.8

If $(a+ib)(c+id)(e+if)(g+ih) = A+iB$, then $\left(a^2+b^2\right)\left(c^2+d^2\right)\left(e^2+f^2\right)\left(g^2+h^2\right) =$

(A) A^2+B^2

(B) A^2-B^2

(C) A^2

(D) B^2

Q.9

If $|z| = \max\{|z-1|, |z+1|\}$ then

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(A) $|z+\bar{z}|=\frac{1}{2}$

(B) $z+\bar{z}=1$

(C) $|z+\bar{z}|=1$

(D) None of these

Q.10

If $z=\frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right)$, then $\left(\frac{|z|}{amp(z)}\right)$ equals

(A) 1

(B) π

(C) 3π

(D) 4

Q.11

If $m_1, m_{2,} m_3$ and m_4 respectively denote the moduli of the complex numbers $1+4i, 3+i, 1-i$ and $2-3i$, then the correct one, among the following is (b) $m_4<m_3<m_2<m_1$

(A) $m_1<m_2<m_3<m_4$

(B) $m_4<m_3<m_2<m_1$

(C) $m_3<m_2<m_4<m_1$

(D) $m_3<m_1<m_2<m_4$

Q.12

For the complex numbers z_1 and z_2 if $\left|1-\bar{z}_1 z_2\right|^2-\left|z_1-z_2\right|^2=k\left(1-\left|z_1\right|^2\right)\left(1-\left|z_2\right|^2\right)$ then ' k ' equals to

(A) 1

(B) -1

(C) 2

(D) -2

Q.13

Let x_1 and y_1 be real numbers. If z_1 and z_2 are complex numbers such that $\left|z_1\right|=\left|z_2\right|=4$, then $\left|\mathrm{x}_1 \mathrm{z}_1-\mathrm{y}_1 \mathrm{z}_2\right|^2+\left|\mathrm{y}_1 \mathrm{z}_1+\mathrm{x}_1 \mathrm{z}_2\right|^2=$

(A) $32\left(\mathrm{x}_1^2+\mathrm{y}_1^2\right)$

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(B) $16\left(x_1^2+y_1^2\right)$

(C) $4\left(x_1^2+y_1^2\right)$

(D) $32\left(x_1^2+y_1^2\right)\left|z_1+z_2\right|^2$

Q.14

Number of solutions of the equation,
$z^3+\frac{3|z|^2}{z}=0$, where z is a complex number and $|z|=\sqrt{3}$ is

(A) 2

(B) 3

(C) 6

(D) 4

Q.15

If $f(z)=\frac{7-z}{1-z^2}$, where $z=1+2i$, then $|f(z)|$ is equal to :

(A) $\frac{|z|}{2}$

(B) $|z|$

(C) $2|z|$

(D) None of these

Q.16

If $\operatorname{Re}\left(\frac{z-1}{z+1}\right)=0$, where $2z=x+$ iy is a complex number, then which one of the following is correct?

(A) $z=1+i$

(B) $|z|=2$

(C) $z=1-i$

(D) $|z|=1$

Q.17

If $2x=3+5i$, then what is the value of $2x^3+2x^2-7x+72$?

(A) 4

(B) -4

(C) 8

(D) -8

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Q.18

If $z^2+z+1=0$, where z is complex number, then the value of $\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+\ldots..+\left(z^6+\frac{1}{z^6}\right)^2$ is

(A) 18

(B) 54

(C) 6

(D) 12

Q.19

$A+iB$ form of $\frac{(\cos x+i\sin x)(\cos y+i\sin y)}{(\cot u+i)(1+i\tan v)}$ is equal to :

(A) $\sin u\cos v[\cos(x+y-u-v)+ i\sin(x+y-u-v)]$

(B) $\sin u\cos v[\cos(x+y+u+v)+ i\sin(x+y+u+v)]$

(C) $\sin u\cos v[\cos(x+y+u+v)- i\sin(x+y-u+v)]$

(D) None of these

Q.20

The minimum value of $|z|+|z-i|$ is

(A) 0

(B) 1

(C) 2

(D) None of these

Answers & Solutions

Q.1 Answer:

4

Solution:

$$(x+iy)^{1/3}=a+ib$$
$$\Rightarrow \quad \left\{(x+iy)^{1/3}\right\}^3=(a+ib)^3$$
$$\Rightarrow \quad x+ty=a^3+3a^2ib+3ai^2b^2+i^3b^3$$
$$\Rightarrow \quad x+iy=\left(a^3-3ab^2\right)+i\left(3a^2b-b^3\right)$$

Comparing the real and imaginary part on both sides,

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$\Rightarrow \quad x = a^3 - 3ab^2$ and $y = 3a^2b - b^3$

$\Rightarrow \quad \frac{x}{a} = a^2 - 3b^2$ and $\frac{y}{b} = 3a^2 - b^2$

$\Rightarrow \quad \frac{x}{a} + \frac{y}{b} = a^2 - 3b^2 + 3a^2 - b^2$

$\Rightarrow \quad \frac{x}{a} + \frac{y}{b} = 4a^2 - 4b^2$

$\Rightarrow$ Hence, $\quad \frac{x}{a} + \frac{y}{b} = 4\left(a^2 - b^2\right)$

Q.2 Answer:

0

Solution:

$x + \mathrm{iy} = \frac{1}{1+\cos\theta+i\sin\theta} = \frac{1}{2\cos^2\frac{\theta}{2}+i2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{1}{2\cos\frac{\theta}{2}} \times$
$\frac{1}{\cos\frac{\theta}{2}+\mathrm{i}\sin\frac{\theta}{2}} = \frac{1}{2\cos\frac{\theta}{2}} \times \frac{1}{\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}}$
$= \frac{1}{2\cos\frac{\theta}{2}} \times \mathrm{e}^{-\mathrm{i}\frac{\theta}{2}} = \frac{1}{2\cos\frac{\theta}{2}}\left(\cos\frac{\theta}{2} - \mathrm{i}\sin\frac{\theta}{2}\right) = \frac{1}{2} - \mathrm{i}\frac{1}{2}\tan\frac{\theta}{2}$
$\Longrightarrow \mathrm{x} = \frac{1}{2}$
$4x^2 - 1 = 4 \times \left(\frac{1}{2}\right)^2 - 1 = 1 - 1 = 0$

Q.3 Answer:

0

Solution:

$\mathrm{z} = (2 - 3\mathrm{i})$
$\mathrm{z}^2 - 4z + 13$
$= (2 - 3\mathrm{i})^2 - 4(2 - 3\mathrm{i}) + 13$
$= -5 - 12\mathrm{i} - 8 + 12\mathrm{i} + 13$
$= -12\mathrm{i} + 12\mathrm{i} - 5 - 8 + 13$
$= 0$

Q.4 Answer:

0

Solution:

$\left(\frac{1+i}{1-i}\right)^{1001} + \left(\frac{1-i}{1+i}\right)^{1001}$
$= \left[\frac{(1+i)^2}{2}\right]^{1001} + \left[\frac{(1-i)^2}{2}\right]^{1001}$
$= \mathrm{i}^{1001} + (-i)^{1001} = 0$

Q.5 Answer:

$n\pi$

Solution:

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$\text{wet } z = \frac{1-i\sin\alpha}{1+2i\sin\alpha}$

$z = \frac{(1-i\sin\alpha)(1-2i\sin\alpha)}{(1+2i\sin\alpha)(1-2i\sin\alpha)}$

$z = \frac{1-2i\sin\alpha-i\sin\alpha+2i^2\sin^2\alpha}{(1)^2-(2i\sin\alpha)^2}$

$z = \frac{1-3i\sin\alpha-2\sin^2\alpha}{1-4i^2\sin^2\alpha}$

$z = \frac{\left(1-2\sin^2\alpha\right)-3i\sin\alpha}{1+4-\sin^2\alpha}$

$z = \frac{1-2\sin^2\alpha}{1+4\sin^2\alpha} - i\frac{3\sin\alpha}{1+4\sin^2\alpha}$

z is purely real so $\text{Im}(z) = 0$

$\frac{3\sin\alpha}{1+4\sin^2\alpha} = 0$

$\Rightarrow \sin\alpha = 0$

$\Rightarrow \alpha = 0, \pi, 2\pi$

$\Rightarrow \alpha = n\pi, n \in I$

Hence answer is $\alpha = n\pi, n \in I$

Q.6 Answer:

2i

Solution:

$\frac{\alpha}{\text{a}} + \frac{\beta}{\text{b}} + \frac{\gamma}{\text{c}} = 1 + \text{i}$ **squaring**

$\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2} + 2\left(\frac{\alpha\beta}{ab} + \frac{\beta\gamma}{bc} + \frac{\gamma\alpha}{ac}\right) = 2i$

or $\frac{\alpha^2}{\text{a}^2} + \frac{\beta^2}{\text{b}^2} + \frac{\gamma^2}{\text{c}^2} + \frac{2\alpha\beta\gamma}{\text{abc}}\left(\frac{\text{c}}{\gamma} + \frac{\text{a}}{\alpha} + \frac{\text{b}}{\beta}\right) = 2\text{i}$

$\therefore \quad \frac{\alpha^2}{\text{a}^2} + \frac{\beta^2}{\text{b}^2} + \frac{\gamma^2}{\text{c}^2} = 2\text{i}$

Q.7 Answer:

$[\pi/2, 3\pi/2]$

Solution:

From conics of complex number,

$|z - 6i| = \text{lm}(z)$ **is a parabola having focus** $6i$ **and directrix as real axis.**

$(\arg z)_{\min} = \frac{\pi}{4}, (\arg z)_{\max} = \frac{3\pi}{4}$

$\frac{\pi}{4} \le \arg z \le \frac{3\pi}{4}$

$\Rightarrow \arg z - \arg \bar{z} = 2\,\text{argz} \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

Q.8 Answer:

$A^2 + B^2$

Solution:

$(a+ib)(c+id)(e+if)(g+ih) = A + iB \quad \ldots$

$\Rightarrow \quad (a-ib)(c-id)(e-if)(g-ih) = A - \text{iB} \quad \ldots$

Multiplying (i) and (ii), we get $\left(a^2+b^2\right)\left(c^2+d^2\right)\left(e^2+f^2\right)\left(g^2+h^2\right) = A^2 + B^2$

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Q.9 Answer:

$|z+\bar{z}|=1$

Solution:

$|z|=\max \text{ of } |z-1|,|z+1|$

Let $z=a+ib$**, where** a **and** b **are real number,** $\therefore |z|=\sqrt{(a-1)^2+b^2}$ **or** $|z|=\sqrt{(a+1)^2+b^2}$ **In the first case,**

$\Rightarrow |z|=\sqrt{(a-1)^2+b^2}=\sqrt{a^2+b^2}$

$\therefore a^2+b^2=(a-1)^2+b^2$

Solving this, we get $a=\frac{1}{2}$

When, $z+\bar{z}=a+ib+a-ib=2a$

$\Rightarrow 2a=2\times\frac{1}{2}=1$

$\Rightarrow |z+\bar{z}|=\sqrt{1^2+0^2}=1$

$\Rightarrow |z+\bar{z}|=1$

Hence, the answer is $|z+\bar{z}|=1$.

Q.10 Answer:

4

Solution:

$z=\frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right)$

$=\frac{\pi}{4}(1+i)^4\left[\frac{1+\pi+\pi+1}{(\sqrt{\pi}+i)(1+\sqrt{\pi}i)}\right]=\frac{\pi}{4}(1+i)^4\frac{2}{i}$

$=\frac{\pi}{4}(2i)^2\frac{2}{i}=2\pi i\ \therefore\left(\frac{|z|}{\text{amp}(z)}\right)=\frac{2\pi}{\frac{\pi}{2}}=4$

Q.11 Answer:

$m_3<m_2<m_4<m_1$

Solution:

Let $z_1=1+4i, z_2=3+i, z_3=1-i$ **and** $z_4=2-3i$

- $m_1=|z_1|, m_2=|z_2|, m_3=|z_3|$ **and** $m_4=|z_4|$

$\Rightarrow \quad m_1=\sqrt{17}, m_2=\sqrt{10}, m_3=\sqrt{2}$**, and** $m_4=\sqrt{13}$,

$\Rightarrow \quad m_3<m_2<m_4<m_1$.

Q.12 Answer:

1

Solution:

$|1-\bar{z}_1z_2|^2-|z_1-z_2|^2$

$=(1-\bar{z}_1z_2)(1-z_1\bar{z}_2)-(z_1-z_2)(\bar{z}_1-\bar{z}_2)$

$=1-z_1\bar{z}_2-\bar{z}_1z_2+z_1\bar{z}_1z_2\bar{z}_2-(z_1\bar{z}_1-z_1\bar{z}_2$

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$-\bar{z}_1 z_2 + z_2 \bar{z}_2)$
$= 1 + z_1 \bar{z}_1 z_2 \bar{z}_2 - z_1 \bar{z}_1 - z_2 \bar{z}_2$
$= 1 + |z_1|^2 |z_2|^2 - |z_1|^2 - |z_2|^2$
$= \left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right)$
$\Rightarrow \quad k = 1$

Q.13 Answer:

$32\left(x_1^2 + y_1^2\right)$

Solution:

$|x_1 z_1 - y_1 z_2|^2 + |y_1 z_1 - x_1 z_2|^2$
$= |x_1 z_1|^2 + |y_1 z_2|^2 - 2\,\mathrm{Re}\left(x_1 y_1 z_1 z_2\right)$
$+ |y_1 z_1|^2 + |x_1 z_2|^2 + 2\,\mathrm{Re}\left(x_1 y_1 z_1 z_2\right)$
$= x_1^2 |z_1|^2 + y_1^2 |z_2|^2 + y_1^2 |z_1|^2 + x_1^2 |z_2|^2$
$= x_1^2 |z_1|^2 + y_1^2 |z_2|^2 + y_1^2 |z_1|^2 + x_1^2 |z_2|^2$
$= 2\left(x_1^2 + y_1^2\right)\left(4^2\right) = 32\left(x_1^2 + y_1^2\right)$

Q.14 Answer:

4

Solution:

$z^3 + \frac{3|z|^2}{z} = 0, \Rightarrow z^3 + \frac{3z \cdot \bar{z}}{z} = 0$
$\Rightarrow z^3 + 3\bar{z} = 0$
Let $z = re^{i\theta}$
$\Rightarrow \quad r^3 e^{i3\theta} + 3re^{-i\theta} = 0$
$\Rightarrow \quad e^{i4\theta} = -1 \quad [\because r = \sqrt{3}]$
$\Rightarrow \cos 4\theta + i \sin 4\theta = -1$
$\Rightarrow \cos 4\theta = -1$
Now $0 \le \theta < 2\pi \Rightarrow 0 \le 4\theta < 8\pi$
$\therefore \theta = \pi, 3\pi, 5\pi, 7\pi$

Q.15 Answer:

$\frac{|z|}{2}$

Solution:

$z = 1 + 2i \Rightarrow |z| = \sqrt{1+4} = \sqrt{5}$
$\therefore f(z) = \frac{7-z}{1-z^2} = \frac{7-1-2i}{1-(1+2i)^2}$
$= \frac{6-2i}{1-(1-4+4i)} = \frac{6-2i}{4-4i} = \frac{3-i}{2-2i}$
$\Rightarrow |f(z)| = \left|\frac{3-i}{2-2i}\right| = \frac{|3-i|}{|2-2i|}$
$= \frac{\sqrt{9+1}}{\sqrt{4+4}} = \frac{\sqrt{5}}{2} = \frac{|z|}{2}$

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Q.16 Answer:

$|z| = 1$

Solution:

$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1}$
$\frac{z-1}{z+1} = \frac{x^2+y^2-1+2iy}{x^2+y^2+2x+1}$
$\Rightarrow \text{Re}\left(\frac{z-1}{z+1}\right) = \frac{x^2+y^2-1}{x^2+y^2+2x+1} = 0$
$\Rightarrow x^2 + y^2 - 1 = 0$
$\Rightarrow x^2 + y^2 = 1$
Also, $z\bar{z} = x^2 + y^2 = 1$
and $z\bar{z} = |z|^2$
$\Rightarrow |z|^2 = 1$
$\Rightarrow |z| = 1$

Q.17 Answer:

4

Solution:

Given $2x = 3 + 5i$
$\Rightarrow \quad x = \frac{3+5i}{2}$
Consider $x^3 = \frac{27+125i^3+225i^2+135i}{8}$
$= \frac{27-125i-225+135i}{8} \quad \left(\because i^2 = -1\right)$
$i^3 = -i$
$= \frac{-198+10i}{8} = \frac{-99+5i}{4}$
and $x^2 = \frac{9+25i^2+30i}{4}$
$= \frac{9-25+30i}{4} = \frac{-8+15i}{2}$
Now, Consider $2x^3 + 2x^2 - 7x + 72$
$= \left(-\frac{99}{2} - 8 - \frac{21}{2} + 72\right) + \left(\frac{5}{2} + 15 - \frac{35}{2}\right) i$
$= \frac{-99-16-21+144}{2} = \frac{8}{2} = 4$

Q.18 Answer:

12

Solution:

$z^2 + z + 1 = 0 \Rightarrow z = \omega$ **or** ω^2
So, $z + \frac{1}{z} = \omega + \omega^2 = -1$
$z^2 + \frac{1}{z^2} = \omega^2 + \omega = -1$
$z^3 + \frac{1}{z^3} = \omega^3 + \omega^3 = 2$
$z^4 + \frac{1}{z^4} = -1, z^5 + \frac{1}{z^5} = -1$ **and** $z^6 + \frac{1}{z^6} = 2$
$\therefore$ **The given sum** $= 1 + 1 + 4 + 1 + 1 + 4 = 12$

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Q.19 Answer:

$\sin u \cos v[\cos(x+y-u-v)+$
$i \sin(x+y-u-v)]$

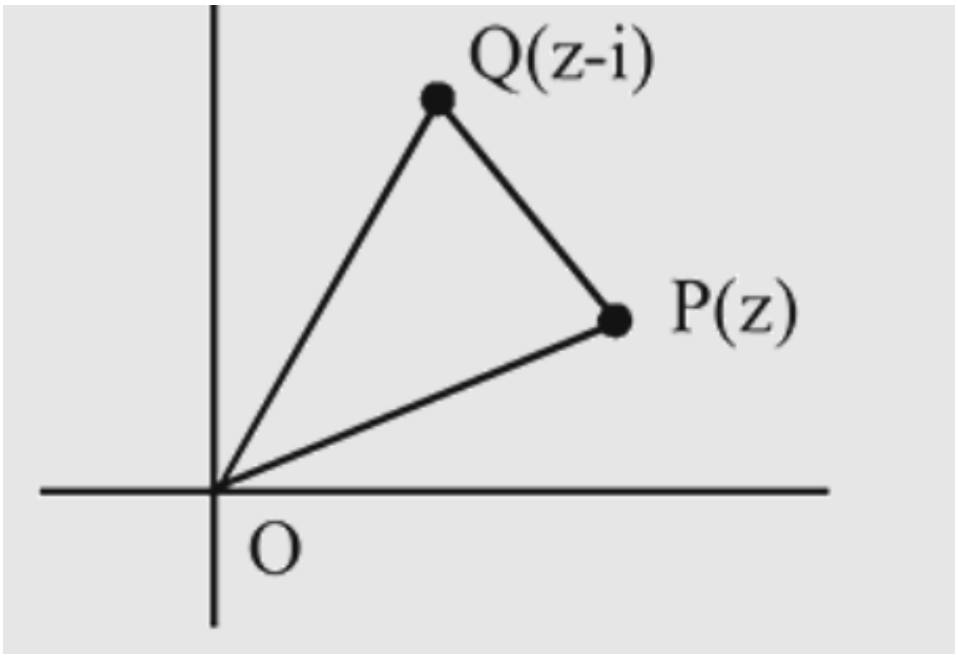
Solution:

Given $\frac{(\cos x+i \sin x)(\cos y+i \sin y)}{(\cot u+i)(1+i \tan v)}$
$= \frac{(\cos x+i \sin x)(\cos y+i \sin y)}{(\cos u+i \sin u)(\cos v+i \sin v)}$
$= \sin u \cos v[\cos(x+y-u-v)$
$+i \sin(x+y-u-v)$

Q.20 Answer:

1

Solution:



Using the result $|z_1+z_2| \leq |z_1|+|z_2|$
we get $|z|+|z-i| = |z|+|i-z|$ [**since** $|z| = |-z| \leq |z+i-z| = |i| = 1$
$\therefore$ **minimum value of** $|z|+|z-i|$ **is 1**

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